Akhilesh Y Ragate

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linkedin.com/in/akhilesh-y-ragate github.com/Quantamaster Portfolio

Professional Summary

AI/ML Engineer with a passion for pushing the limits of what can be done with AI. and hands-on experience in computer vision, audio signal analysis, remote sensing, and data engineering. Proven ability to build and deploy end-to-end AI systems using CNNs, YOLO, SAM, and cloud-based ETL pipelines. Strong problem-solving background with specialize in creating innovative solutions that leverage into emerging tech. .

Education

Don Bosco Institute of Technology Bangalore, India
Bachelor of Engineering in Computer Science Specialization in (AI & ML) CGPA-8.5 *2020 – 2024*

Skills

Programming: Python, C, C++, Java, SQL, JavaScript,R
AI / ML: TensorFlow, Keras, Scikit-Learn, OpenCV, CNNs, YOLO,SAM,XGBoost
Data Engineering: Pandas, PySpark, ETL/ELT, Data Warehousing
Cloud & Tools: AWS, Oracle Cloud, Docker, PowerBI, FastAPI, Git ,GCP,Android Studio,Hugging Face,chroma DB , Supabase,Gemini Api,AI Studio
Databases: MySQL, Oracle DB, Postgresql
Operating System: Windows, Linux, Android, RTOS, CLI

Experience

AIML & Applications Development Intern Bangalore
Planet Eye Infra AI (Mitcon Engineering)

- Working on AI/ML-based applications including image ,audio processing,remote sensing and model deployment.
- Developing an ”ML-driven satellite imagery” analysis system to detect truck layby zones along highways, delivering geospatial visualizations and deployable inference pipelines.
- Developed an”AI/ML-based NDT solution” for detecting wall hollowness and internal defects using acoustic & ultrasonic signals, leveraging CNN/CNN-LSTM models and advanced audio feature engineering;currently involved in **Assessment Management**.
- Enabling **early defect detection, reduced manual inspection effort, improved structural safety**.
- Asset Management project “NHIT Road Distress Detection ” states using AI&ML for detection of highway infrastructure, mainly developed pothole distress detection using algorithms for web and mobile app.
- Developed a application using the SAM model that achieved a 95% accuracy rate in detecting potholes, overlaying masks and storing latitudinal and longitudinal within PDF reports,improving traceability of field observations.

Python Data Engineer Trainee Intern Bangalore
Linkzie Technologies

- Built data pipelines using Python, SQL, and cloud services.Project ”Real-Time Charging Station Utilization and Performance Analytics.” Built a serverless ETL pipeline on AWS using S3, Lambda, Glue, Athena, and Parquet
- Track utilization by location and time.Identify failure patterns or maintenance needs from error logs.Optimize grid load, session pricing, and user experience with near-real-time insights.Prepare aggregated, partitioned datasets for dashboarding, predictive analytics, and reporting using Athena.
- Implemented ETL workflows and optimized data processing.Project”E-commerce Product Data Aggregator and Analyzer”,enabling centralized product aggregation and insightful analytics through RESTful APIs
- Implemented data cleaning, transformation, and schema unification to handle heterogeneous vendor data formats.Enabled insightful analytics such as price comparison, availability tracking, and category-wise trends.Improved data accessibility and decision-making by providing a scalable, API-driven analytics layer.

Projects

Kidney Stone Detection — AI/ML(Tensorflow, Keras , CNN Neural network) Main Project

- Developed a CNN-based model to detect kidney stones from medical images.
- This project we use technique that is used to find out kidney stones from 5mm to 10inchs in diameter which is using neural networks.
- Implemented using TensorFlow and Keras for accurate classification.

Missile Guidance System—Digital Image Processing,Computer Vision,Neural Networks

- Implemented ANN-based object detection using YOLO and COCO datasets.
- Performed missile detection on static images using Python and OpenCV.
- Using Artificial Neural Networks (ANN) along with YOLO and COCO configuration (with Python) to perform a missile detection on a static image.

Stock Price Predictor— Machine Learning (SVM, Random Forest,Logistic Regression)

- Built ML models using historical stock market data.
- Applied Logistic Regression, SVM, and Random Forest techniques. Technical Analysis by means of statistics such as past price and volume and apply on machine learning technique which develop a financial data predictor program.
- This contains the datasets storing all historical stocks prices and data will be as input for training set model of the program to provide precise stock price.

Exam Preparation App — Android Application Development

- To develop a strategy-based exam preparation app on subject wise preparation methods and implement testing methods to reduce inaccuracy by computerized schedule.
- Automated Study Scheduling: Eliminates manual planning by generating computerized, subject-wise schedules, reducing human error and improving preparation consistency.
- Scalable Architecture: Modular Android app design allows easy addition of new subjects, tests, and features .

Online Voting System — Database Management Systems

- Developed a secure voting platform using PHP frontend and SQL backend.
- To implement specific task as to vote a candidate from a list ,elect or determine the result by using PHP as Front-end and SQL as Back-end in Database management system.

Certifications

Oracle - Cloud Infrastructure 2025 Data Science Professional
Oracle - Cloud Infrastructure 2025 Generative AI Professional
Microsoft - Azure Fundamentals with Java Fundamentals
Udemy - Quantum Computing , Skill Nation – Data Analysis using AI
Cisco Networking Academy - Python Essentials
Responsible AI: Applying AI Principles with Google Cloud

LANGUAGES

English - Full Professional Proficiency
Marathi - Full Professional Proficiency
Hindi - Full Professional Proficiency
Kannada - Full Professional Proficiency

INTERESTS

Playing and watching Cricket , chess, and football; listening to music, blogs, content creation & tech podcasts.

Achievements & Activities

- **Computer Society of India (CSI) Member:** Participated in technical workshops and events focused on emerging technologies, contributing to continuous learning and professional development.
- Completed **150+ coding challenges** on **LeetCode, HackerRank, and GeeksforGeeks**.
- Participated in **GenAI hackathons and technical quizzes** (Tata Crucible, Hack GenAI, GUVI)